

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397838
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Shipman; Gregory Dean

Cart

Abstract

Disclosed is a cart comprising a retaining frame assembly configured to support and retain a door in a vertical position, the retaining frame assembly comprising a base support frame and side support members coupled to the base support frame. In certain embodiments, the cart may further comprise a wheel assembly including a front wheel and a rear wheel assembly, wherein the front wheel assembly comprises a single swivel wheel coupled to a bottom side of the retaining frame assembly, wherein the rear wheel assembly comprises a pair of wheels coupled to opposing outer sides of the retaining frame assembly, and wherein the wheels of the rear wheel assembly are larger than the wheel of the front wheel assembly.

Inventors:	Shipman; Gregory Dean (Richland, WA)
Applicant:	Shipman; Gregory Dean (Richland, WA)
Family ID:	1000006887240
Appl. No.:	18/088567
Filed:	December 25, 2022

Publication Classification

Int. Cl.: B62B3/10 (20060101); B62B3/12 (20060101); B62B5/00 (20060101)

U.S. Cl.:

CPC B62B3/108 (20130101); B62B3/12 (20130101); B62B5/0083 (20130101);

Field of Classification Search

CPC: B62B (3/12); B62B (3/108); B62B (5/0083)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3731947	12/1972	Fontaine	182/127	E06C 1/397
4545592	12/1984	Taskovic	280/35	A45C 13/385
8662486	12/2013	Holder	269/152	B25B 1/00
10017199	12/2017	Silva	N/A	B62B 3/02
10696315	12/2019	Monk	N/A	A01K 29/00

Primary Examiner: Evans; Bryan A

Attorney, Agent or Firm: Plager Schack LLP

Background/Summary

BACKGROUND

(1) The present disclosure relates generally to carts for transport of planar items such as doors.
(2) Transporting a planar item such as a door on a construction site and/or other areas, particularly where the terrain may be rough, can be challenging and inconvenient. As such, an improved transport system is desirable.

SUMMARY

(3) According to various embodiments, disclosed is a cart comprising: a retaining frame assembly configured to support and retain a door in a vertical position, the retaining frame assembly comprising a base support frame and side support members coupled to the base support frame; a wheel assembly configured to rollably support the retaining frame assembly, the wheel assembly comprising a front wheel assembly coupled to a front section of the retaining frame assembly, and a rear wheel assembly coupled to a rear section of the retaining frame assembly, the rear wheel assembly and the front wheel assembly, wherein the front wheel assembly comprises a single swivel wheel coupled to a bottom side of the retaining frame assembly, wherein the rear wheel assembly comprises a pair of wheels coupled to opposing outer sides of the retaining frame assembly, and wherein the wheels of the rear wheel assembly are larger than the wheel of the front wheel assembly.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The detailed description of some embodiments of the invention will be made below with reference to the accompanying figures, wherein the figures disclose one or more embodiments of the present invention.
(2) FIG. 1 is a side perspective view of a cart in accordance with various embodiments.
(3) FIG. 2 is a front elevation view of the cart.
(4) FIG. 3 is an exploded perspective view of the cart.
(5) FIG. 4 is a top plan view of the cart.
(6) FIG. 5 is a side perspective view, similar to FIG. 1, showing the cart in use for transporting a door.
(7) FIG. 6 is a bottom detail perspective view of a front wheel assembly of the cart.

(8) FIG. 7 is a side elevation view of the cart in use.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

(9) According to various embodiments as depicted in FIGS. 1-7, disclosed is a cart **10** configured for wheeling a door **11** or other planar object over various terrain. In certain embodiments, cart **10** (also referred to as “door cart **10**”) may generally comprise a base support frame **14** and side support members **20**, which together form a retaining frame assembly **10A** configured to support door **11** within cart **10**. In certain embodiments, door cart **10** further comprises a front wheel assembly **22** and a rear wheel assembly **24** coupled to retaining frame assembly **10A** for rollably supporting retaining frame assembly **10A**. In further embodiments, door cart **10** may also include a cord **12** which may be coupled to retaining frame assembly **10A**.

(10) In embodiments, base support frame **14** of retaining frame assembly **10A** may comprise a longitudinal bar **14A** and a transverse bar **14B**, wherein transverse bar **14B** may be coupled to longitudinal bar **14A** proximate a rear end (i.e., at a rear section) of longitudinal bar **14A** and in cross alignment thereto. In certain embodiments, transverse bar **14B** may be slightly offset from the rear end of longitudinal bar **14A** as shown in the figures. In further embodiments, transverse bar **14B** may be coupled to a bottom side of longitudinal bar **14A**, such that longitudinal bar provides a substantially leveled bottom support surface for door **11** or other item being transported on cart **10**, while transverse bar **14B** may support rear wheel assembly **24**.

(11) In some embodiments, longitudinal bar **14A** and transverse bar **14B** are elongated aluminum tubes but are not limited to this option. According to an exemplary embodiment, longitudinal bar **14A** and transverse bar **14B** may have a square cross area that is approximately 2 inches by approximately 2 inches, with wall thickness of approximately $\frac{1}{8}$ of an inch. Longitudinal bar **14A** may be approximately 42 inches long, while transverse bar **14B** may be approximately 16 inches long. It shall be appreciated however, that bars **14A** and **14B** may be made of various materials and may be of various sizes, dimensions, and geometric configurations in alternate embodiments.

(12) In some embodiments, side support members **20** may comprise front vertical supports **20A** and rear vertical supports **20B**. In certain embodiments, front vertical supports **20A** and rear vertical supports **20B** may each comprise two planar brackets configured to retain door **11** from the sides. In some embodiments, front vertical supports **20A** may comprise a pair of brackets coupled on right and left sides (i.e. opposing outer sides) of longitudinal bar **14A** proximate a front end of the bar.

(13) In certain embodiments, rear vertical supports **20B** may comprise a pair of brackets coupled on right and left sides of longitudinal bar **14A** proximate a rear end of the bar. In some embodiments, vertical supports **20B** may situate over transverse bar **14B** and may include bottom openings **20C** configured to frame over the top and sides of bar **14B** as shown. In some further embodiments, a pair of triangular brackets **16** may be attached to a top side of transverse bar **14B** on outer sides of vertical supports **20B** (i.e., between vertical supports **20B** and rear wheels **24**), and are configured to brace supports **20B**. In certain embodiments, rear vertical supports **20B** may be higher than front vertical supports **20A**. According to an exemplary embodiment, both front vertical supports **20A** and rear vertical supports **20B** may be made from a plastic material, and may be about $\frac{3}{4}$ of an inch in thickness, with a general width of about 4 inches. Additionally, front vertical supports may be about 6 inches long, while rear vertical supports may be about 20 inches long. It shall be appreciated however, that front vertical supports **20A** and rear vertical supports **20B** may be made of various materials and may be of various sizes, dimensions, and geometric configurations in alternate embodiments.

(14) In certain embodiments, cushion blocks **18** may be provided on various sections of longitudinal bar **14A** for cushioning door **11** or other item placed atop longitudinal bar **14A**. In one embodiment, one cushion block **18** may be placed proximate the rear end of longitudinal bar **14A** (i.e., between rear vertical supports **20B**) and two cushion blocks **18** may be placed proximate the front end of longitudinal bar **14A** (i.e., between front vertical supports **20A**), as shown in FIG. 4.

Each block **18** may be made from a rubber material and may be about $\frac{5}{8}$ of an inch in thickness, about 2 inches in width, and about $1\frac{1}{4}$ inches in length. It shall be appreciated that various cushioning materials, which may be of different sizes, dimensions, and/or geometric configurations may be used in alternate embodiments, and that cushion blocks **18** may be omitted in alternate embodiments.

(15) In certain embodiments, door cart **10** may comprise 3 wheels in total, wherein front wheel assembly **22** may comprise a single swivel caster wheel **22A** (“front wheel **22A**”) positioned directly beneath longitudinal bar **14A**, and rear wheel assembly **24** may comprise two large rear wheels **24A** positioned to the sides of longitudinal bar **14A**. This wheel configuration enables cart **10** to maneuver easily around corners, up or down inclines, and/or over rough terrain, including debris, gravel, ruts, cords, etc. In one embodiment, the outer diameter ratio between front wheel **22A** and rear wheels **24A** may be between about 0.3 to about 0.4 but is not limited to this option. According to an exemplary embodiment, front wheel **22A** may have an outer diameter of approximately 7 inches, while rear wheels **24A** may each have an outer diameter of about 18 inches.

(16) In certain embodiments, front wheel **22A** may be attached to a bottom side of longitudinal bar **14A** via a swivel wheel bracket **22B**. Swivel wheel bracket **22B** may be attached to a bottom side of longitudinal bar **14A** via a swivel attachment pin assembly **22C** engaged through a bottom hole within longitudinal bar **14A**. Front wheel **22A** may be rotationally coupled to swivel wheel bracket **22B** via an axial bolt **22D** engaged through a corresponding pair of holes **22D** at a bottom section of swivel wheel bracket **22B**. In embodiments, several hole pairs **22D** may be provided along the bottom section of swivel wheel bracket **22B** to enable height adjustment of wheel **22A**. As such, front wheel **22A** is positioned underneath longitudinal bar **14A** and is configured to swivel for allowing easy maneuverability of cart **10**. In some embodiments, back wheels **24A** may be coupled to transverse bar **14B** via a pair of L-brackets **19**. In embodiments, each L-bracket **19** may be attached to a side end of transverse bar **14B**, wherein wheel **24A** may be rotationally coupled to an outer side of the L-bracket via an axial pin assembly **19A**.

(17) In certain embodiments, cord **12** may be configured to enable a user to pull cart **10** and/or wrap around door **11** placed atop the cart (see FIG. 7). In one embodiment, cord **12** may be attached at a front end of longitudinal bar **14A**. In certain embodiments, cord **12** may be coupled to the swivel pin in swivel attachment pin assembly **22C**, which centers cord **12** with respect to longitudinal bar **14A**, cart **10**, for providing a centered pull force line. In some further embodiments, cord **12** may be looped, enabling it to easily be wrapped around door **11** and/or be pulled by a user. It shall be appreciated that one or more pull cords **12** may be coupled to other parts of cart **10** in alternate embodiments, and that pull cord **12** may be omitted in some alternate embodiments.

(18) It shall be appreciated that the components of cart **10** may be assembled using any known techniques in the field. In one embodiment, components of cart **10** including longitudinal bar **14A**, transverse bar **14B**, front vertical supports **20A**, rear vertical supports **20B**, triangular brackets **16**, cushion blocks **18**, components of wheel assemblies **22**, **24**, may be attached via an assembly of nuts, washers, and bolts engaged through corresponding holes within the components as shown in the figures. It shall be appreciated however, that different attachment mechanism as known in the art may be used in alternate embodiments.

(19) In use, door **11** or other object may be placed atop longitudinal bar **14A**, between vertical supports **20A**. In case of transport of a door with hinges, the door may lay sideways on the cart with the hinge side positioned upwards. The cart may then be transported over most terrains to deliver the door to its designated location. The front of the cart may then be tipped up to allow the door to slide out of the cart, in an upright position for installation.

(20) As such, the disclosed subject matter provides a cart that enables a user to easily move doors over challenging terrain and obstacles efficiently and safely and to position the door properly for

installation. The retaining frame assembly **10A** formed by base support frame **14** and side support members **20** is configured to securely support a door in a vertical position rather than a flat planar position for transport. This reduces the door's footprint, making it less awkward to transport, and enables it to easily be positioned for installation. Additionally, as the rear and front ends of the frame assembly are unobstructed, it can support a door of any length, and allows the door to be placed with its major axis in horizontal alignment, for reduced height and greater stability in transport.

(21) It shall be appreciated that cart **10** may be used for transporting doors of various sizes and configurations, and further for transporting other planar objects such as planks, beams, sidings, windows, mirrors, pictures, and the like.

(22) It shall be appreciated that cart **10** can have multiple configurations in different embodiments. It shall be appreciated that the components of cart **10** may comprise any alternative known materials in the field and be of any size and/or dimensions. It shall be appreciated that the components of cart **10** described herein may be manufactured and assembled using any known techniques in the field. Furthermore, the components of cart **10** may be provided to the end user as an assembled and/or integral unit, or as a kit/assembly of parts according to alternate embodiments.

(23) As used herein, the articles “a” and “an” are intended to include one or more items, and may be used interchangeably with “one or more.” Where only one item is intended, the term “one” or similar language is used. Also, as used herein, the terms “has”, “have”, “having”, “with” or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

(24) The constituent elements of the disclosed device and system listed herein are intended to be exemplary only, and it is not intended that this list be used to limit the device of the present application to just these elements. Persons having ordinary skill in the art relevant to the present disclosure may understand there to be equivalent elements that may be substituted within the present disclosure without changing the essential function or operation of the device. Terms such as ‘approximate,’ ‘approximately,’ ‘about,’ etc., as used herein indicate a deviation of within $\pm 10\%$. Relationships between the various elements of the disclosed device as described herein are presented as illustrative examples only, and not intended to limit the scope or nature of the relationships between the various elements. Persons of ordinary skill in the art may appreciate that numerous design configurations may be possible to enjoy the functional benefits of the inventive systems. Thus, given the wide variety of configurations and arrangements of embodiments of the present invention the scope of the invention is reflected by the breadth of the claims below rather than narrowed by the embodiments described above.

Claims

1. A cart for transporting a door comprising: a retaining frame assembly configured to support and retain said door in a vertical position, the retaining frame assembly comprising a longitudinal bar and side support members coupled to the longitudinal bar; a wheel assembly configured to rollably support the retaining frame assembly, the wheel assembly comprising a front wheel assembly coupled to a front section of the longitudinal bar, and a rear wheel assembly coupled to a rear section of the longitudinal bar, wherein the front wheel assembly comprises a single swivel wheel coupled to a bottom side of the retaining frame assembly, wherein the rear wheel assembly comprises a pair of wheels coupled to opposing outer sides of the retaining frame assembly, and wherein the wheels of the rear wheel assembly are larger in diameter than the wheel of the front wheel assembly, wherein the side support members comprise front vertical supports and rear vertical supports, wherein the front vertical supports comprise a pair of planar brackets coupled to opposing sides of the longitudinal bar proximate the front wheel assembly, wherein the rear vertical supports comprise a pair of planar brackets coupled to opposing sides of the longitudinal bar

proximate the rear wheel assembly, wherein the retaining frame assembly is configured to support said door, such that the door is positioned atop the longitudinal bar with the door's major vertical axis in alignment with the longitudinal bar and wherein the door is bracketed between the pair of planar brackets of the front vertical supports and the pair of planar brackets of the rear vertical supports, and wherein the retaining frame assembly is unobstructed at its rear and front ends.

2. The cart of claim 1, wherein a diameter ratio of the wheel of the front wheel assembly to the wheels of the rear wheel assembly is between about 0.3 to about 0.4.
3. The cart of claim 1, further comprising a pull cord coupled to the front section of the retaining frame assembly.
4. The cart of claim 1, further comprising a transverse bar which is shorter than the longitudinal bar, the transverse bar being coupled to a rear section of the longitudinal bar at a bottom side of the longitudinal bar, and in cross alignment to the longitudinal bar.
5. The cart of claim 4, wherein the rear wheel assembly is coupled to opposing outer ends of the transverse bar, and wherein the front wheel assembly is coupled to a front section of the longitudinal bar and at the bottom side of the longitudinal bar.
6. The cart of claim 4, wherein the pair of planar brackets of the rear vertical supports are vertically higher than the pair of planar brackets of the front vertical supports.
7. The cart of claim 4, further comprising a first cushioned block attached to a top side of the longitudinal bar at the front section of the longitudinal bar, and a second cushioned block attached to the top side of the longitudinal bar at the rear section of the longitudinal bar.
8. The cart of claim 4, wherein the longitudinal bar and the transverse bar each comprise tubular bars having a square or rectangular cross section.
9. The cart of claim 5, further comprising a pair of triangular brackets attached to a top side of transverse bar and configured to brace the pair of planar brackets of the rear vertical supports.
10. The cart of claim 1, further comprising: a transverse bar coupled to a rear section of the longitudinal bar in cross alignment to the longitudinal bar, the pair of wheels in the rear wheel assembly being coupled to opposing outer ends of the transverse bar; and a pair of triangular brackets attached to a top side of transverse bar, wherein each triangular bracket in the pair of triangular brackets is disposed between a respective one of the pair of planar brackets of the rear vertical supports and a corresponding wheel of the rear wheel assembly, wherein the pair of triangular brackets are configured to brace the pair of planar brackets of the rear vertical supports.
11. The cart of claim 1, wherein the swivel wheel of the front wheel assembly is coupled to the bottom side of the longitudinal bar via a swivel wheel bracket, the swivel wheel bracket being attached to a bottom side of the longitudinal bar via a swivel attachment pin engaged through a bottom hole in the longitudinal bar and the swivel wheel is coupled to the swivel wheel bracket via an axial bolt engaged through a corresponding pair of holes in the swivel wheel bracket.
12. The cart of claim 1, further comprising a pull cord coupled to the swivel wheel attachment pin, wherein the pull cord is centered with respect to the longitudinal bar for providing a centered pull force line on the cart, and wherein the cord is configured to wrap around said door when said door is supported on the retaining frame assembly.
